# Analysis of Copper Pieces

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# Outline

# Goals of the Report

- **Explore Patterns:** Identify relationships among chemical elements.
- **Data Simplification:** Reduce complexity through dimensionality reduction.
- **Identify Groupings:** Uncover natural clusters in the data.
- **Gain Insights:** Understand how elemental concentrations vary across observations and zones.

# Dataset Overview

- **Dataset Details:**
  - 95 observations with concentrations of chemical elements.
  - Elements include Cu, Ag, As, Co, Fe, Ni, Pb, Sb, Sn.
  - Categorical variable **"Local1"** identifies 7 zones: C, D, H, J, P, S, U.
- **Data Characteristics:**
  - No missing values (all variables have 95 non-null entries).
  - Units not specified (e.g., ppm, $\mu$g/g, mg/L are typical).
- **Variable Types:**
  - ID (Integer), Local1 (Categorical), Elemental Concentrations (Float/Integer).

# Dataset Preview

- Example of the first 6 rows of the dataset:

| ID | Local1 | Cu | Ag | As | Co | Fe | Ni | Pb | Sb | Sn |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | J | 75 | 0.08 | 4.07 | 0.03 | 5.25 | 9.72 | 0.04 | 0.02 | 0.01 |
| 2 | D | 81 | 0.01 | 1.42 | 1.14 | 8.62 | 3.79 | 0.03 | 0.14 | 0.02 |
| 3 | D | 85 | 0.03 | 2.26 | 0.06 | 5.61 | 1.83 | 0.03 | 4.10 | 0.01 |
| 4 | D | 87 | 0.03 | 7.96 | 0.37 | 3.13 | 0.40 | 0.08 | 1.67 | 0.02 |
| 5 | D | 83 | 0.03 | 8.46 | 0.34 | 2.05 | 0.18 | 0.12 | 1.17 | 0.09 |
| 6 | D | 85 | 0.03 | 7.14 | 0.36 | 4.07 | 0.18 | 0.10 | 0.24 | 0.05 |

Table: First 6 rows of the dataset.

# Exploratory Data Visualization

- **Histograms and Boxplots:**
  - Provide insights into distribution and variability.
  - Histograms reveal the shape of the data distribution.
  - Boxplots summarize key statistics (median, IQR, outliers).

# Histogram Observations

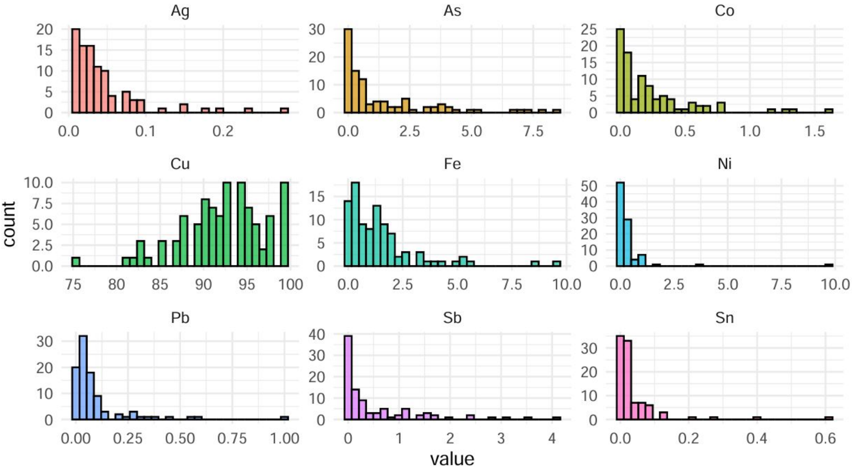



Figure 1: Histograms of the variables

# Boxplots Observations

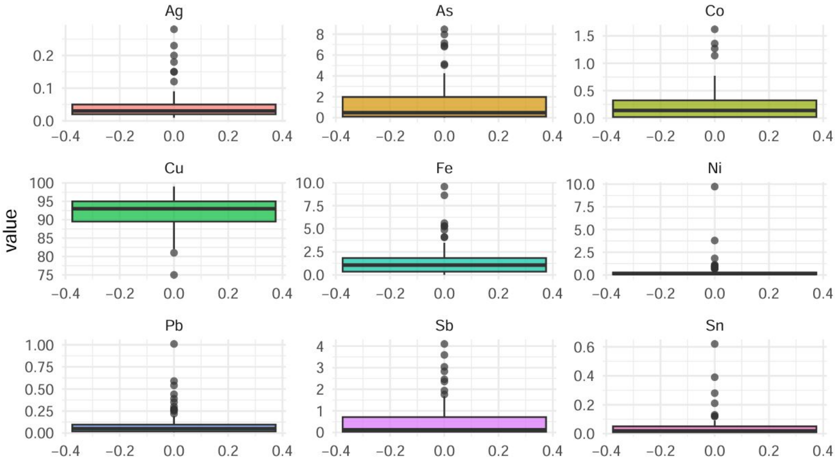



Figure 2: Boxplots of the variables

# Correlation and Covariance Analysis

- It's necessary to analyze the **covariance** and **correlation** between the variables.
- The variables exhibit different variability as seen in **boxplots and histograms**.
- When the mean and standard deviations differ significantly among variables, it's better to use the **correlation matrix** instead of the covariance matrix.

# Correlation Matrix

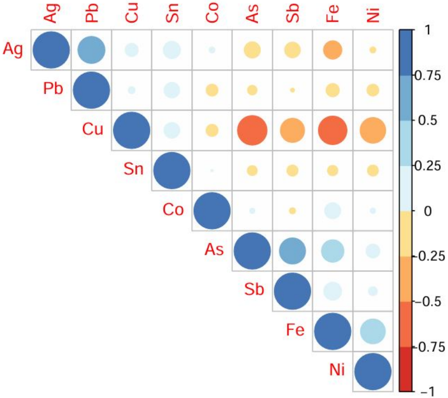



Figure 3: Correlation Matrix

- Observations:
  - Cu is **highly negatively correlated with As, Sb, Fe, and Ni**.
  - As is **positively correlated with Sb and Fe**.
  - Fe is **positively correlated with Ni**.

# Standard Deviation of Variables

| Element | Standard Deviation |
|---|---|
| Cu | 4.799496 |
| Ag | 0.0474584 |
| As | 1.945533 |
| Co | 0.311733 |
| Fe | 1.685588 |
| Ni | 1.077465 |
| Pb | 0.1463703 |
| Sb | 0.8351631 |
| Sn | 0.0813433 |

- Standard deviations are not consistent in size.
- Additionally, the means are also **not comparable**.
- Therefore, it's more appropriate to perform PCA using the **correlation matrix** rather than the covariance matrix.

# Variance Explained by Components

**Key Insights:**

- When we look at the eigenvalues, we retain components with eigenvalues greater than 1 (**Kaiser Criterion**).
- We observed that the first 3 components have eigenvalues greater than 1 and explain about 61.9% of the total variance.

| Component | Std. Dev. | Proportion of Variance | Cumulative Proportion |
|---|---|---|---|
| Comp.1 | 1.6875583 | 0.3164281 | 0.3164281 |
| Comp.2 | 1.2348125 | 0.1694180 | 0.4858461 |
| Comp.3 | 1.0945272 | 0.1331100 | 0.6189561 |
| Comp.4 | 0.9961741 | 0.1102625 | 0.7292186 |
| Comp.5 | 0.9358784 | 0.0973187 | 0.8265373 |

# Scree Plot Analysis

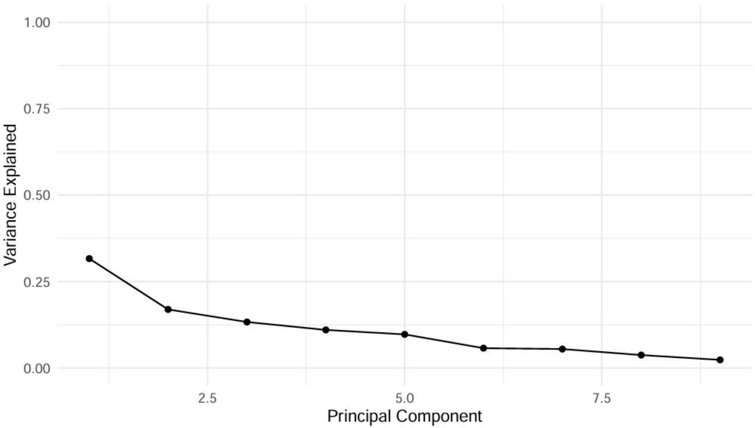



Figure 4: Scree Plot of the variance explained

- The scree plot shows a sharp decline after the first 2 components.

# Variance Explained by Components

**Why 3 Components?**

- The 3 components capture a significant portion of variability in the dataset.
- Including a 4th component doesn't contribute much additional information according to our scree plot analysis.
- Combining all our observations, it seems that the most suitable choice for our case is the choice of 3 main components.

**Reliable Solutions with Mean and Scree Plot Criteria**

Require at least one of the following conditions:

- Number of **variables** $< 30$ (**satisfied**)
- Number of **cases (individuals)** $> 250$ (**not satisfied**)

# Calculation of the Threshold Rule

**Threshold Rule for Significant Contributions:**

$$\text{Threshold} = \sqrt{\frac{\lambda}{p}}$$

- $\lambda$: Eigenvalue of a PC (explained variance).
- $p$: Total number of variables in the dataset.

**Thresholds and Most Important Variables for First 3 PCs:**

| PC | Threshold | Significant Variables |
|---|---|---|
| PC1 | 0.5625 | Cu, As, Fe, Sb |
| PC2 | 0.4116 | Ag, Pb |
| PC3 | 0.3648 | Co, Ni, Sb |

# Contribution of Each Variable to a PC

**Formula for Contribution ($a_{ij}^2$):**

$$a_{ij}^2 = \left(\frac{l_{ij}}{\lambda_j}\right)^2$$

- $l_{ij}$: Loading of variable $X_i$ on PC $Y_j$.
- $\lambda_j$: Eigenvalue of $Y_j$ (variance explained by the PC).

**Example Contributions:**

- PC1: Cu (0.2625), As (0.1998), Fe (0.1891), Sb (0.1186).
- PC2: Ag (0.4204), Pb (0.3560).
- PC3: Co (0.2654), Ni (0.2402), Sb (0.2372).

**Interpretation:**

- For PC2: Pb explains 0.42%, Ag explains 0.35%.
- Contributions help determine the key variables defining each PC.

# Correlation Between Variables and PCs

**Formula for Correlation ($r_{ij}$):**

$$r_{ij} = a_{ij} \cdot \sqrt{\lambda_j}$$

- $r_{ij}$: Correlation between variable $X_i$ and PC $Y_j$.
- $a_{ij}$: Contribution of $X_i$ to $Y_j$.
- $\lambda_j$: Eigenvalue of $Y_j$.

**Correlations for PCs:**

- PC1: Cu (0.8646), As ($-0.7339$), Fe ($-0.7545$), Sb ($-0.5811$).
- PC2: Ag ($-0.7368$), Pb ($-0.8006$).
- PC3: As (0.3776), Co ($-0.5331$), Ni ($-0.5364$), Sb (0.5639).

**Interpretation:**

- High correlations indicate strong relationships with the PC.
- Variables with low correlations contribute less to defining the PC.

# PCA Biplot

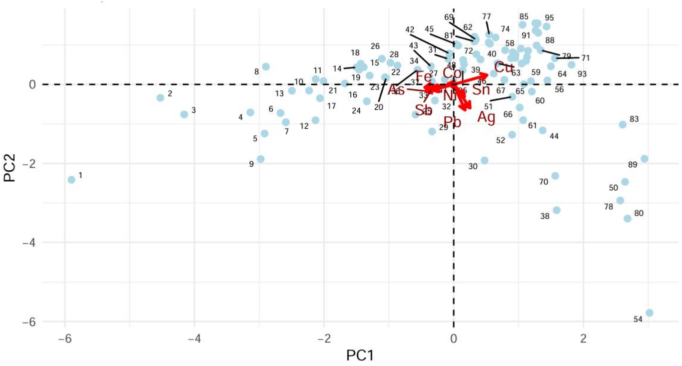



Figure 5: Plot of the first 2 Principal Components

- Each light blue dot represents a sample projected onto the 2-dimensional space of PC1 and PC2.
- PC1 (horizontal axis) and PC2 (vertical axis) explain most of the variability in the data.

# Cluster Analysis Overview

## Cluster Analysis: A Method of Grouping Data

- Grouping statistical units (individuals, objects, or variables) into **similar groups**.
- Objects within the same group are **more similar to each other** than to those in different groups.
- It is an important method for:
  - **Exploratory analysis**
  - Reducing the **dimensionality of data**

## Why Normalize Data?

- Means and variances of the variables are on **different scales**.
- Without normalization, variables with higher variance (e.g., Cu) would dominate the analysis.
- Normalization (e.g., **z-scores**) ensures all variables contribute **equally to Euclidean distance calculations**.

# Euclidean Dissimilarity Matrix

**Head of the Dissimilarity Matrix**

| Observation | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.00000 | 52.60047 | 83.86473 | 92.94495 |
| 2 | 52.60047 | 0.00000 | 42.05705 | 43.12171 |
| 3 | 83.86473 | 42.05705 | 0.00000 | 22.27001 |
| 4 | 92.94495 | 43.12171 | 22.27001 | 0.00000 |

- Numbers represent the **Euclidean distances between observations**.
- For example:
  - Distance between Observations 1 and 2 is 52.6.
  - Distance between Observations 4 and 5 is only 2.39, indicating higher similarity.

# Hierarchical Clustering - Single Linkage Method

**Single Linkage Overview**

- In agglomerative hierarchical clustering, the single linkage method defines the distance between clusters as the **shortest distance** between any pair of observations from different clusters.
- The merge matrix shows the indices of observations or clusters merged at each stage of the hierarchical process.
- The agglomerative coefficient (**0.934**) is high, indicating that clusters are well-defined and cohesive.
- Single linkage tends to form **"chain-like"** clusters, which may link outliers to distinct clusters, potentially causing distortions, especially in noisy datasets.
- Further validation can be done by comparing it with other methods like complete linkage or average linkage.

# Hierarchical Clustering (Single Linkage Method)

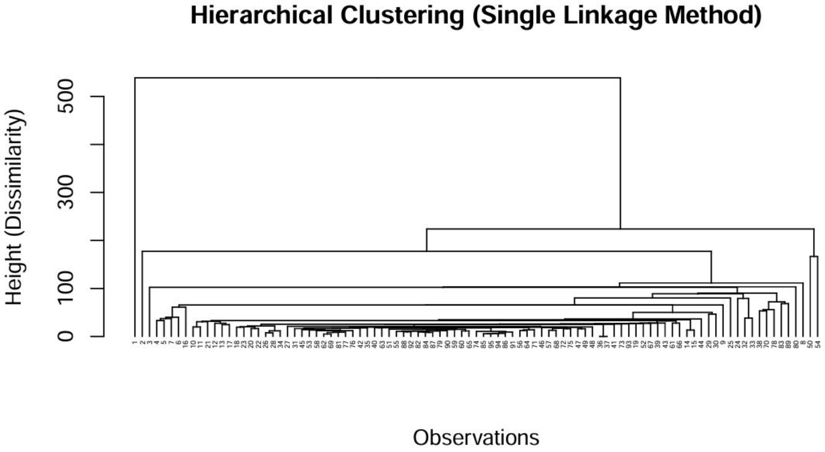



Figure 6: Hierarchical Clustering (Single Linkage Method)

# Hierarchical Clustering: Complete Linkage Method

**Complete Linkage Overview**

- Distance between clusters is defined as the **maximum distance** between any two points from each cluster.
- Produces **compact and evenly distributed clusters**.
- Less sensitive to chaining effects or outliers compared to single linkage.
- Agglomerative coefficient: **0.9532** (higher than single linkage), indicating a stronger clustering structure.
- Dendrogram shows clear hierarchical merges, with distinct clusters forming at various height levels.

# Hierarchical Clustering (Complete Linkage Method)

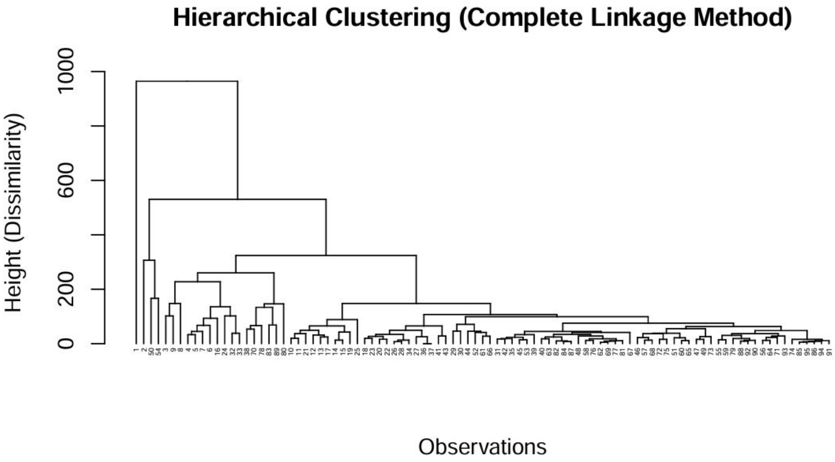



Figure 7: Hierarchical Clustering (Complete Linkage Method)

# Hierarchical Clustering: Ward's Method

### Ward's Method Overview

- Minimizes total variance within clusters, producing **compact and less sensitive clusters**.
- Agglomerative coefficient: **0.9672** (highest among methods), indicating exceptionally strong clustering.
- Preferred for creating **interpretable and cohesive clusters**.
- Dendrogram shows clusters merging at progressively larger dissimilarity levels.
- However, the cophenetic coefficient is low (**0.6197**), suggesting distortion of original pairwise distances.

# Hierarchical Clustering (Ward Linkage Method)

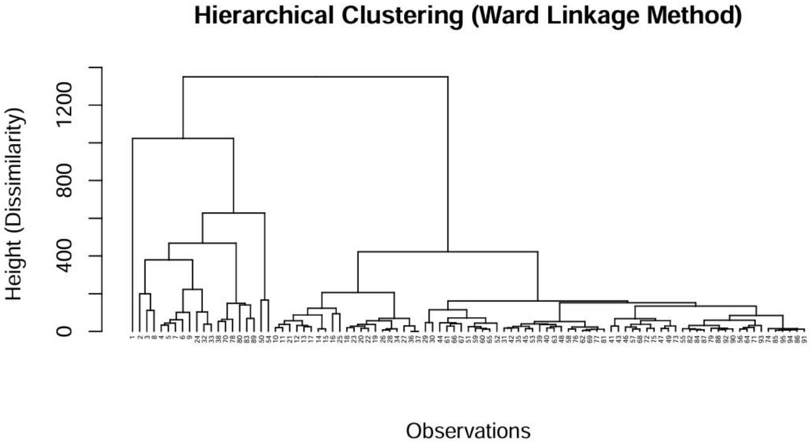



Figure 8: Hierarchical Clustering (Ward Linkage Method)

# Comparison of Methods

**Agglomerative Coefficient (AC):**

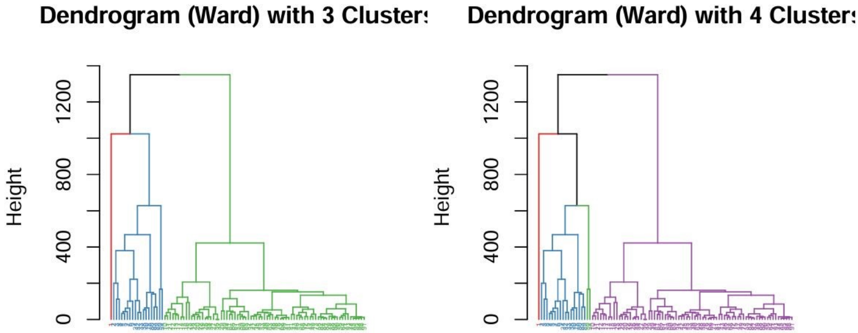
- Single Linkage: **0.9343**
- Complete Linkage: **0.9532**
- Ward's Method: **0.9672** (highest, best clustering structure).

**Cophenetic Coefficient (CC):**

- Single Linkage: **0.8747** (best at preserving original distances).
- Complete Linkage: **0.8515**.
- Ward's Method: **0.6198** (low, distorts pairwise distances).

**Conclusion:** Despite the lower CC, Ward's Method is preferred for its strong clustering structure and minimized variance.

# Dendrogram Comparison: 3 vs. 4 Clusters



# Optimal Cluster Analysis with Ward's Method

**Key Insights:**

- The dendrogram suggests **3 or 4 clusters** as the most natural separations.
- **Dendrogram Heights (Top Jumps):**

| Jump Interval | Height Difference |
|---|---|
| 1350.53 – 1023.45 | 327.08 |
| 1023.45 – 627.86 | 395.59 |
| 627.86 – 467.92 | 159.94 |

- **Conclusion:** 3 clusters appear optimal, with Observation 1 standing out as an outlier.

# PCA vs. Cluster Analysis Summary

**Principal Component Analysis (PCA):**

- Identified **three primary components** explaining 61.9% of total variance:
  - **PC1**: Dominated by Copper (Cu), inversely related to Arsenic (As), Antimony (Sb), and Iron (Fe).
  - **PC2**: Differentiates samples based on Silver (Ag) and Lead (Pb).
  - **PC3**: Captures variability driven by Cobalt (Co) and Nickel (Ni).
- Enabled dimensionality reduction for easier interpretation.

**Cluster Analysis:**

- **Hierarchical clustering (Ward's method):** Produced a **3-cluster solution** with an agglomerative coefficient of 0.97.
- **Key findings:**
  - Aligns with PCA groupings.
  - Highlights a single outlier cluster, warranting further investigation.

**Synergy and Insights:**

- PCA highlighted variability-driving dimensions; clustering revealed natural groupings.

# Findings and Applications of PCA and Clustering

**PCA Insights:**

- PC1 highlights distinct **Cu-rich zones**, potentially due to metallurgical processes, contamination, or geological formations.
- PC2 isolates areas enriched in **Ag and Pb**, useful for environmental impact assessments or tracing industrial waste.

**Clustering Insights:**

- Cluster-based analysis reveals local-to-cluster relationships (e.g., zones P and H contain only Cluster 3 samples).
- Can aid in provenance detection for new samples or identifying potential element relationships.

**Applications and Limitations:**

- Provides a framework for provenance prediction of unknown samples.
- Requires expert validation to confirm or refute assumptions on geology and element interactions.

# References


- Carrasquinha, Eunice (2024). *Análise de Dados Multivariados (ADM)* [Lecture and tutorial slides]. FCUL ULisboa. Available at: https://moodle.ciencias.ulisboa.pt/course/view.php?id=5600
- R Core Team (2024). *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing, Vienna, Austria. https://www.r-project.org/
- **R Packages Used:**
  - dendextend, laGP, purrr, factoextra, cluster, RColorBrewer, metan, corrplot, ggplot2, readxl, tidyverse, dplyr, tidyr, knitr